

D.I.R.T.T. by Dirty Dawgs

Nolan Brechtel (Project Manager), George Utz (Chief Scientist),
Cara Belew, Zachary Morrow, Rebecca Ueltschey
Department of Electrical and Computer Engineering
Mississippi State University



Problem and Target User

- Approximately 25,000 people, including more than 10,000 children, die every day due to starvation [1]
- Agricultural imports in the U.S. exceed its exports by \$41 billion each year [2]
- 49% of U.S. crop fields struggle with poor health and resource concerns like nutrient imbalances [3]

Previous Solutions

- **ROGO Soil Sampling Robot**
- **Agrobot Lala**



- **Large, requires fuel to operate**
- **Expensive and is typically rented**
- **Requires lab area set up to test soil**

Competitive Advantages

Inexpensive

- Price is less than \$1,000

Small and Compact

- Can easily be stored when out of use

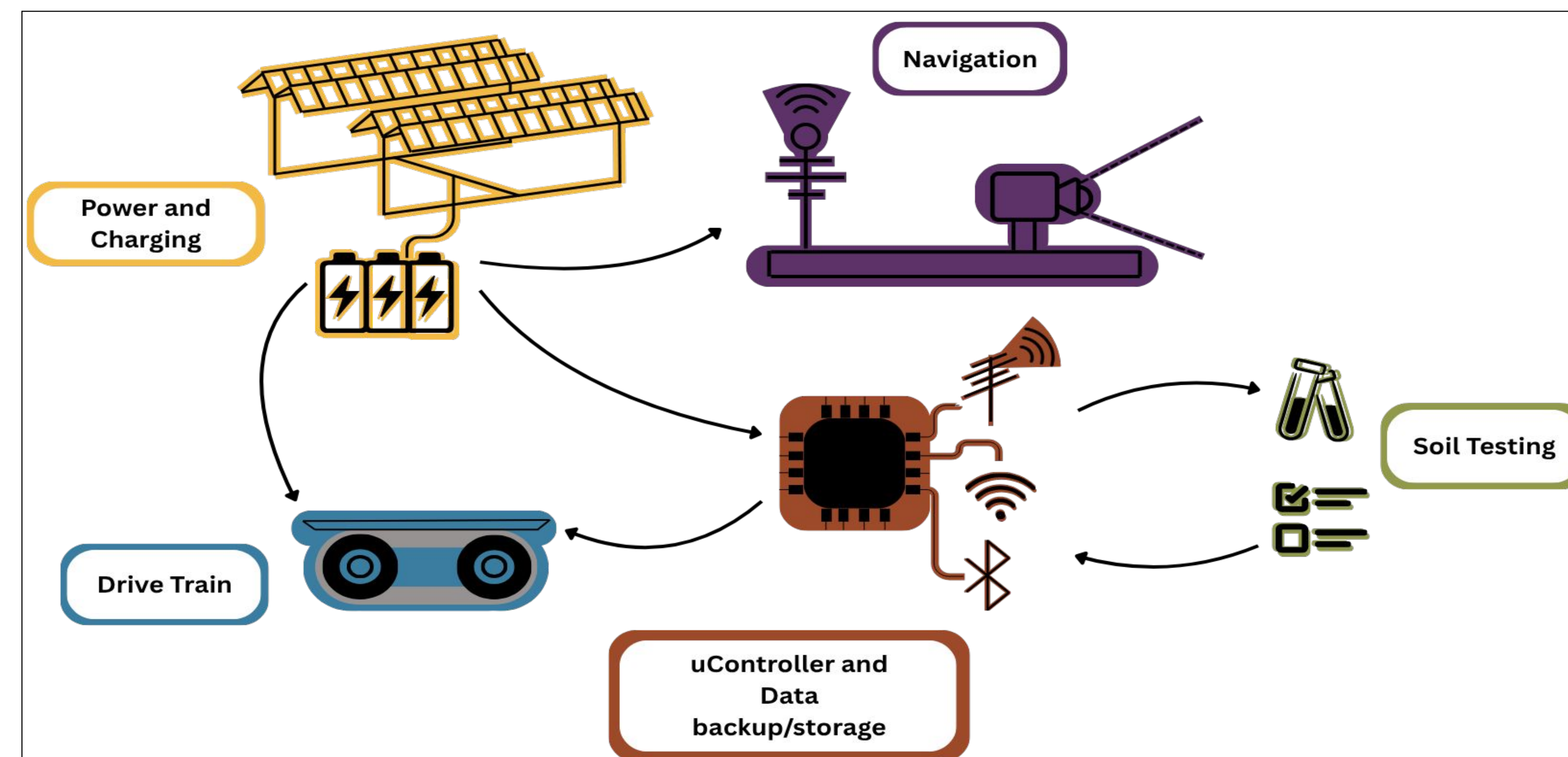
Solar Powered

- No need for gas or any other power source

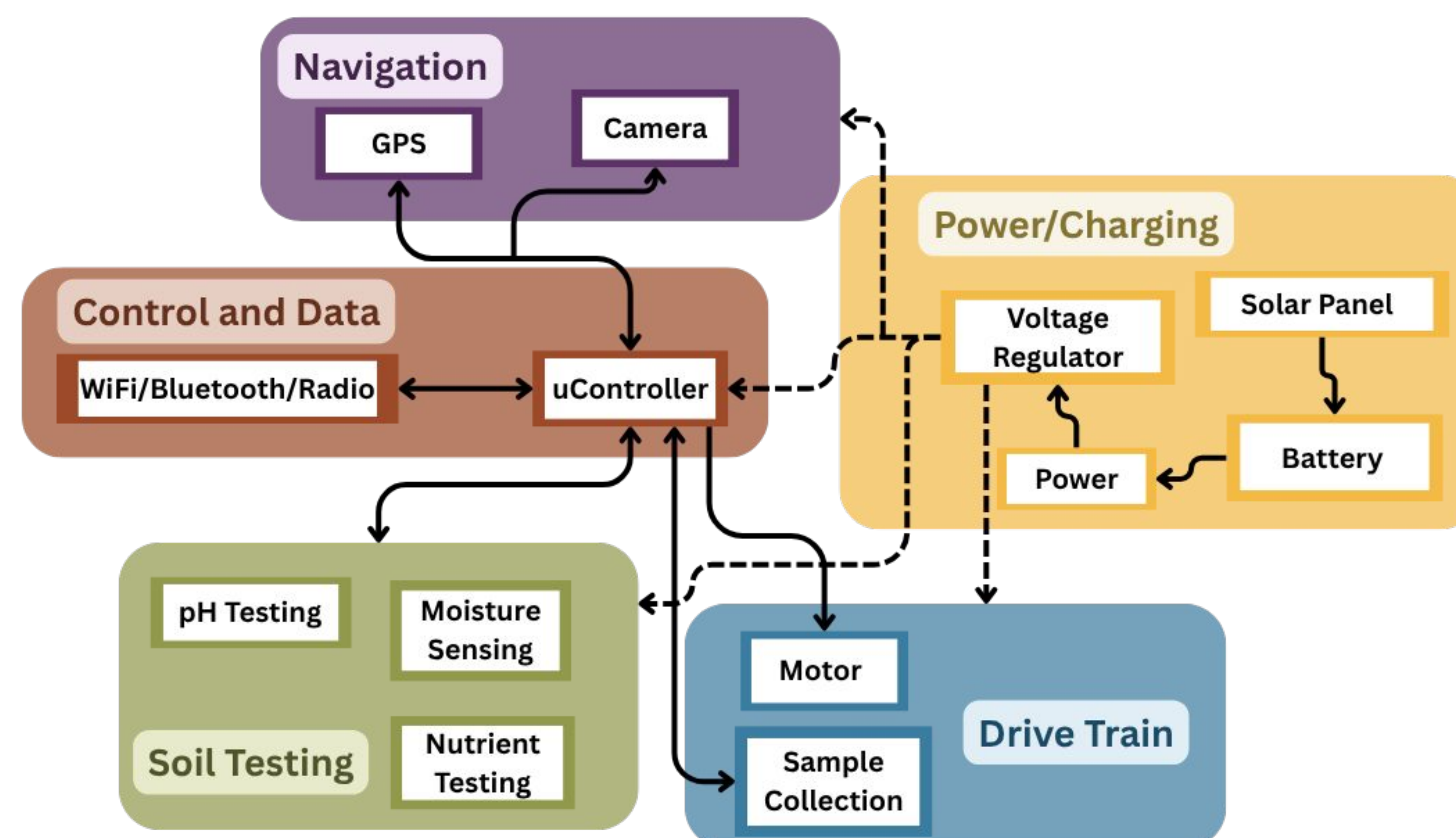
Self-Contained

- Can test soil without needing to go to a set lab

Solution (Simple Infographic)



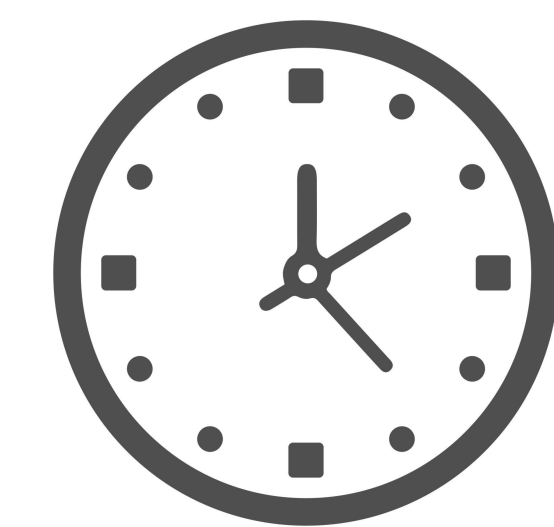
System Diagram



Broader Impact

Customer Impact

- Allows for testing both more quickly and around the clock



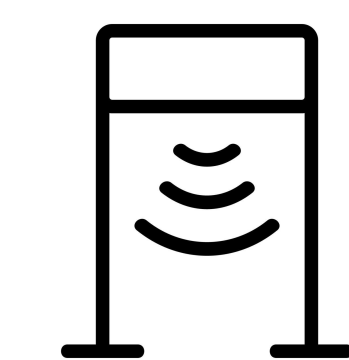
Broader Impact

- Gives farmers the tools to make better food



Future Expansion

- Swappable sensor to test for whatever users may want



References

1. J. Holmes. "Losing 25,000 to Hunger Every Day." [un.org](https://www.un.org/). Accessed: Sep. 9, 2026. [Online.] Available: [Losing 25,000 to Hunger Every Day | United Nations](https://www.un.org/)
2. J. Kaufman. "Ag and Food Statistics Charting the Essentials." ers.usda.gov. Accessed: Sep. 9, 2026. [Online.] Available: [Ag and Food Statistics: Charting the Essentials - Agricultural Trade | Economic Research Service](https://ers.usda.gov/)
3. A. B. Rosenberg and S. Wallander "Farmers Report Soil Related Resource Concerns on About Half of Soybean Wheat Cotton and Oat Fields." ers.usda.gov. Accessed: Sep. 9, 2026. [Online.] Available: [Farmers Report Soil-Related Resource Concerns on About Half of Soybean, Wheat, Cotton, and Oat Fields | Economic Research Service](https://ers.usda.gov/)